

Technical Note

CATALYST - PLASTICIZER PRODUCTION

DESCRIPTION

Dorf Ketal Tyzor® organic titanates are valuable, environmentally friendly catalysts for the direct esterification of aromatic and aliphatic mono, di, tri, and other carboxylic acids to make a variety of esters, some of which are used as plasticizers.

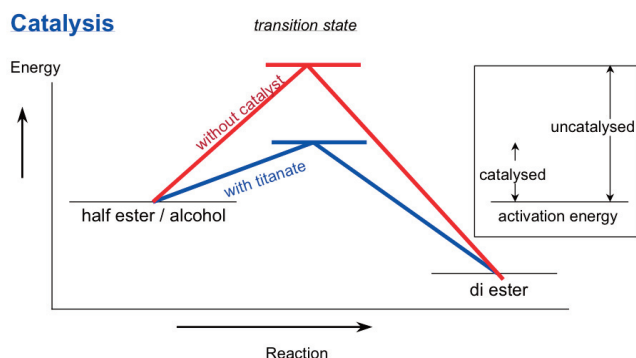
Plasticizers are important polymer additives that improve processing and enhance desirable product properties such as flexibility and toughness. Although epoxidized vegetable oils, phosphates and polymeric plasticizers are used, most plasticizers are diesters of phthalic acid that contain alkoxy groups such as butylates, n-octylates, iso-octylates and 2-ethylhexylates used to modify the properties of polyvinyl chloride (PVC).

The most common example is di-2-ethyl hexyl phthalate (also called Di Octyl Phthalate, abbreviated DOP). This phthalate is typically prepared by esterification of phthalic anhydride (PA) with corresponding alcohols, but other commonly used di- and tricarboxylate plasticizers can also be made by esterification of acids with alcohols.

Esterification of phthalic anhydride and other di- and tricarboxylic acids to plasticizers requires an esterification catalyst, and Tyzor® organic titanate catalysts are popular choices due to their many performance advantages.

Graph 1 illustrates how titanates reduce activation energy, enhancing reaction control under moderate reaction conditions with few side reactions.

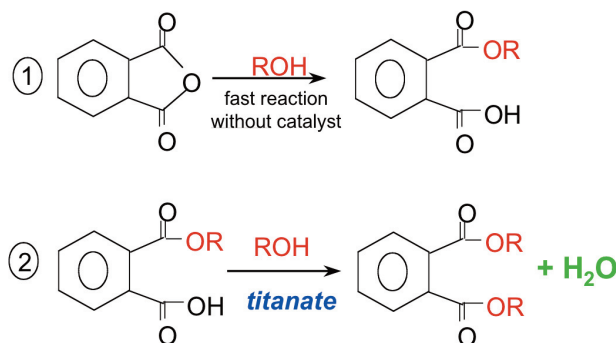
Graph 1: Catalytic Effect of Titanates



PREPARATION OF DI OCTYL PHTHALATE (DOP) WITH TITANATES

DOP is prepared in a two-step reaction of phthalic anhydride with 2-ethyl hexanol.

Figure 1: DOP Synthesis Reaction Scheme



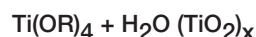
Reaction step 1 is a fast exothermic addition under ring opening and formation of the half ester that occurs at about 130 - 150 °C (266 - 302 °F).

Step 2 is an esterification in which titanates act as a catalyst between the half ester and the alcohol. The water byproduct must be eliminated to avoid saponification of the ester and to force the reaction to the diester (by azeotrope distillation and separation).

TYZOR CATALYST PRODUCTS AND PROPERTIES

Dorf Ketal offers several Tyzor® titanates for plasticizer production. Product selection requires an understanding of their chemical behavior.

a) Titanates are very reactive with water or moisture and readily form polymeric titanium oxides. As little as 1 lb of water can react with 30 lbs of titanate catalyst.



Titanates other than Tyzor® LA, which is very stable in water, must be handled in dry environments.

b) Titanates quickly undergo alcohol exchange.



Basic titanates such as Tyzor® TPT or TnBT and their mixtures will quickly form corresponding higher titanates by transesterification with the alcohol in the reaction mixture.

Handling

Due to their high moisture sensitivity, titanates must be stored and handled under appropriate conditions:

- Exclusion of moisture
- Use of dry ingredients
- Addition as final reaction component

Purging with nitrogen and use of drying tubes are recommended.

The low melting point of Tyzor® TPT (approximately 18 °C [64.4 °F]) must be considered in cool or cold climates, but crystallized product can be dissolved without loss of performance.

TITANATE CATALYST SELECTION CONSIDERATIONS

When selecting titanates for use as esterification catalysts it is helpful to consider temperature, titanate type, catalyst amount, discoloration potential and the types of products to be produced.

Temperature

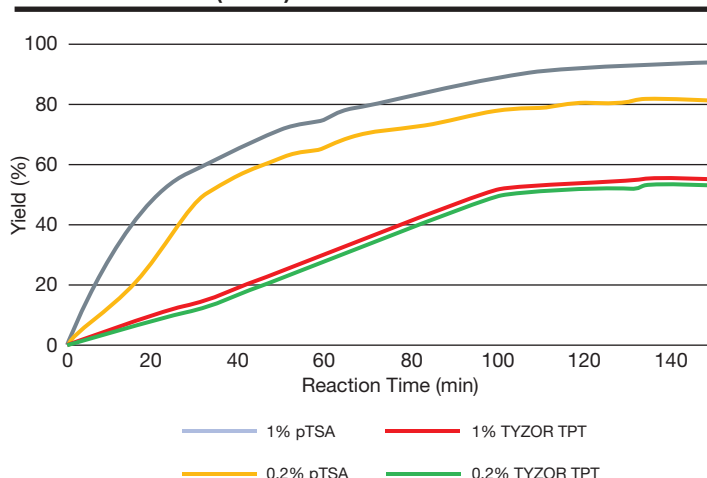
Titanates used in esterification reactions for plasticizer production require reaction temperatures ranging from 180 °C (356 °F) to 220 °C (428 °F). By way of contrast, transesterifications readily run at temperatures below 100 °C (212 °F).

This can be seen in Graph 2 when reacting phthalic anhydride with 2-ethylhexanol at 150 °C (302 °F) and in Graph 3 when the reaction is run at 180-220 °C (356-428 °F). At 150 °C (302 °F) the reaction catalyzed by titanates is slow and uncompleted, but at 180-220 °C (356-428 °F) the reaction proceeds rapidly and yields nearly 100%. The graphs also show that protic acids such as p-toluene sulfonic

acid are superior in esterification reactions to Lewis acids at lower temperatures.

This characteristic effectively limits low-pressure use of titanates to esters with alcohols having boiling points of approximately 180 °C (356 °F). However, pressurized equipment can be used for alcohols with lower boiling points.

Graph 2: Esterification of PA with 2-Ethylhexanol @ 150 °C (302 °F)



Graph 3: Esterification of PA with 2-Ethylhexanol @ 180 - 220 °C (356 - 428 °F)

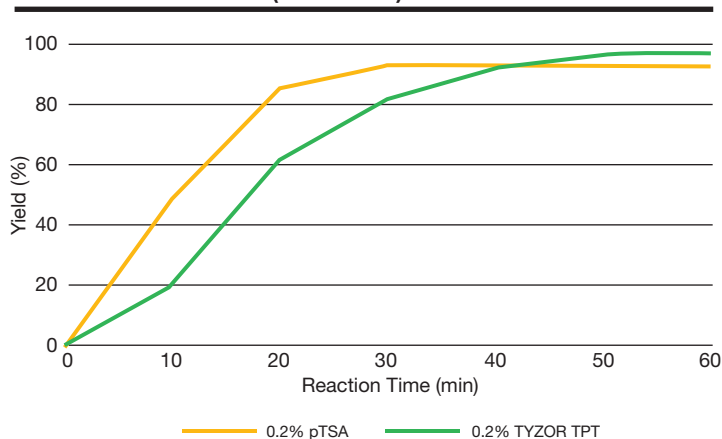


Table 1: Tyzor® Titanates for Plasticizer Production

PRODUCT	FORMULA	MOL WEIGHT	TiO ₂ CONTENT	REMARKS
Tyzor® TPT	Ti(O-i-C ₃ H ₇) ₄	284	27 %	mp = 18 °C very moisture sensitive
Tyzor® TnBT	Ti(OC ₄ H ₉) ₄	340	24 %	very moisture sensitive
Tyzor® BTP	C ₄ H ₉ O-[Ti(OC ₄ H ₉) ₂] _n C ₄ H ₉	1000-1500	34.5 %	viscosity : ca 4000 mPa*s
Tyzor® TPT-20B	preparation of Tyzor® TPT (80%) Tyzor® TnBT (20%)		27 %	very moisture sensitive
Tyzor® TOT	Ti(OC ₈ H ₁₇) ₄	565	14.0 %	less moisture sensitive
Tyzor® TIOT	Ti(O-i-C ₈ H ₁₇) ₄	565	14.0 %	less moisture sensitive
Tyzor® LA	lactic acid/ammonium salt		14.0 %	water stable titanate

A list of alcohols commonly used for plasticizer production follows with their boiling points.

Table 2: Alcohol Boiling Point Data

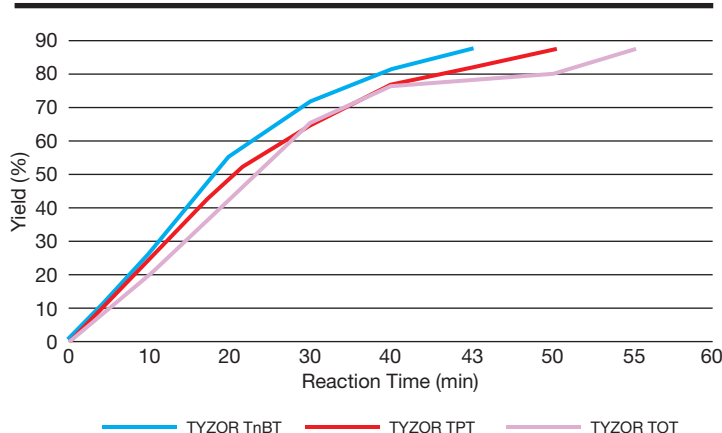
TYPE	BOILING POINT °C (°F)
Methanol	65 (149)
Ethanol	78 (172.4)
Butanol	117 (242.6)
Isobutanol	108 (226.4)
n-Hexanol	156 (312.8)
Isoheptanol	158 - 160 (316.4 - 320)
n-Octanol	194 (381.2)
2-Ethyl hexanol	186 (366.8)
Isooctanol	184 - 190 (363.2 - 374)
n-Nonanol	205 (401)
Isononanol	193 - 202 (379.4 - 395.6)
Isodecanol	215 - 222 (419 - 431.6)
Undecanol	245 (473)
Dodecanol	255 (491)
Tridecanol	276 (528.8)
Benzyl alcohol	205 (401)

Titanate Type

The nature of the alkoxy group R in various titanates with the general structure $Ti(OR)_4$ does not significantly influence reaction rate. The decisive factor for the performance and reactivity of a titanate grade is the TiO_2 content. Tyzor® TPT and TPT/TnBT mixtures are highly effective because they contain the highest TiO_2 content. This can be seen in Graph 4 where Tyzor® TnBT, TPT and TOT are compared at the same molar ratio.

Selecting a titanate with the same alcohol used for the esterification plasticizer (e.g. Tyzor® TOT for the preparation of DOP) avoids the use of mixed alkoxy groups.

Graph 4: Effect of Titanate Type on Reactivity
0.7 mol% Esterification of PTA with 2-Ethylhexanol

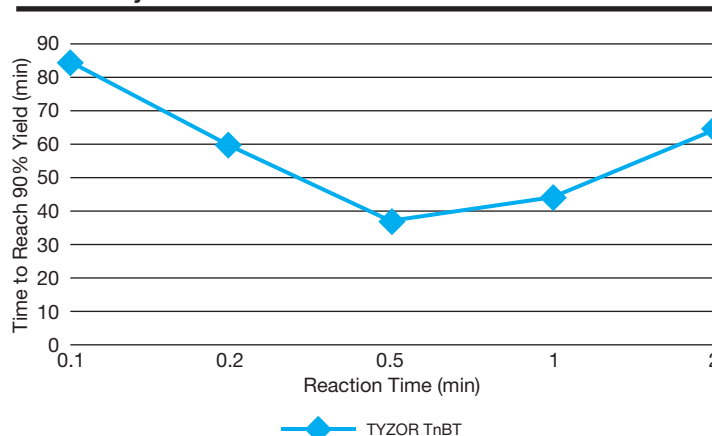


Titanium chelates show remarkably lower activity in esterification reactions. Corresponding zirconates of the structure $Zr(OR)_4$ are not sufficiently reactive to catalyze esterification reactions for plasticizers.

Catalyst Amount

The amount of titanate required for catalysis is very low - typically between 0.05 and 0.5 percent. Graph 5 illustrates the influence of increased Tyzor® TnBT dosages on the time required to reach a 90% yield in the reaction of 1 mole of PA with 3 moles of 2-ethyl hexanol.

Graph 5: Effect of Catalyst Amount on Reaction Rate
Tyzor® TnBT



Discoloration

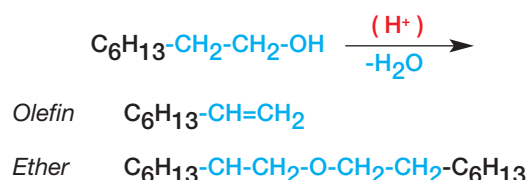
Plasticizer discoloration is typically caused by:

- Oxidation
- Formation of byproducts such as olefins (Titanates strongly reduce olefin formation.)
- Yellowing caused by titanates

Discoloration may be removed easily by filtration in the presence of absorbing additives and other means.

Byproduct Formation

Titanates are very soft catalysts. By way of comparison, protic catalysts are defined as hard acids that catalyze the dehydration of alcohols to form olefins or ethers.



Negative effects of these undesired reactions are:

- Yield reduction referred to the alcohol
- Alcohol recovery, recycling and work-up problems
- Alcohol rearrangement
- Discoloration
- Cost
- Product quality

Secondary alcohols are very sensitive to dehydration. As can be seen in Table 3, titanates prevent alcohol decomposition in the reaction of phthalic anhydride with octanol-2.

Table 3: Esterification of PA with Octanol-2 (50% excess alcohol)

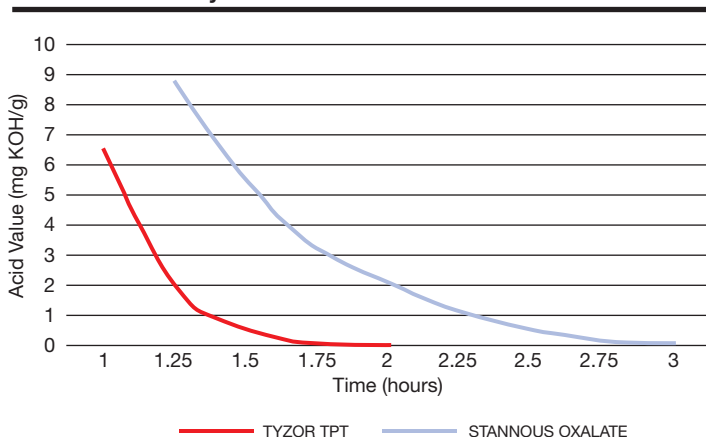
CATALYST	RX TIME	YIELD OF DIESTER	DEHYDRATION
None	37 hours	90%	10%
P-TSA	6 hours	98%	92%
Tyzor® TPT	6.5 hours	99%	5%

Table 3 shows that the reaction rate is very low without catalyst. The protic catalyst p-toluene sulfonic acid (p-TSA) increases the reaction rate significantly but produces a high level of undesirable alcohol degradation. This occurs via protonation of the alcohol OH group and dehydration / olefin formation. Titanates don't cause this elimination and work quickly to produce high diester yields without alcohol dehydration.

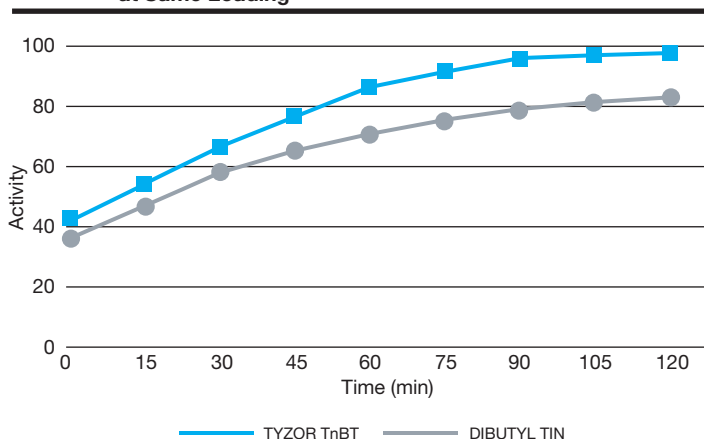
COMPARISON WITH TIN COMPOUNDS

As can be seen in Graphs 6 and 7, Tyzor® titanates are even more reactive than the tin compounds used for many years as Lewis Acid catalysts in esterifications to produce plasticizers. The reaction in Graph 6 was performed at 200 °C (392 °F)/0.2% catalyst (calc. on acid)/20% excess alcohol.

Graph 6: Catalytic Activity of Tyzor® TPT versus Stannous Oxalate in DOP Synthesis



Graph 7: Tyzor® TnBT versus Dibutyl Tin Catalyst at Same Loading



Organotin catalysts have a number of other disadvantages:

- High to extreme toxicity
- Lower reactivity (longer reaction time, higher acid value)
- Complete catalyst removal is difficult
- Catalyst disposal can be environmentally hazardous

CONCLUSION

Tyzor® titanates are excellent catalysts for plasticizer manufacturing that offer decisive advantages over other catalyst systems:

- High product quality
- Reduced waste
- Easy processibility
- Lower cost

Advantages like these have made Tyzor® organic titanates the industrial standard in plasticizer production.

Dorf Ketal offers a range of Tyzor® titanates designed specifically for plasticizer production, and a variety of plasticizers other than dioctyl phthalate (DOP) can be produced with titanate catalysts.

Contact Dorf Ketal today to learn more about the advantages of Tyzor® titanates in your processes.



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